Juan Carlos Gadea Brenes

Data Engineer & Full Stack Developer | Technical Lead, Solutions Architect & DBA

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PROFESSIONAL SUMMARY

Data Engineer and Full Stack Developer with 5+ years of experience, also working as Technical Lead, Solutions Architect and DBA. I design and implement Medallion data architectures (Bronze-Silver-Gold), Data Warehouses and ETL/ELT pipelines on SQL Server (SSIS), PostgreSQL and Oracle, together with advanced analytics platforms (Python, Power BI) for regulated sectors such as banking, government, insurance and telecommunications. I administer, monitor and tune Oracle 19c databases on Oracle Database Appliance (RAC, Data Guard, ASM) and PostgreSQL, with a focus on query tuning and execution-plan analysis. As a Full Stack Developer, I build enterprise and SaaS applications with C# .NET 8, React, Next.js, Angular and Node.js, deployed on Azure, AWS and GCP with CI/CD and FinOps practices, and I apply generative AI (RAG, OCR, LLMs via API) to data extraction and retrieval. Experienced in leading agile squads (Scrum/Kanban), managing executive stakeholders and contributing to bid processes. Currently pursuing a Master's in Big Data & Data Engineering (UCM, 2026-2027). Based in Rome, with daily-use B2 Italian; available for remote or hybrid projects in Italy, the EU and LATAM.

PROFESSIONAL EXPERIENCE

Technical Lead & Solutions Architect | Data Engineer & DBA (Software Application Developer)

NEXUS CONSULTORÍA & SERVICIOS | Remote | Nov 2025 - Present

- **Data Engineering & Data Warehousing:** Design and implementation of scalable data ecosystems under the Medallion architecture (Bronze-Silver-Gold) on SQL Server to improve data quality and analytical traceability. ETL/ELT pipelines orchestrated with SSIS and Python scripts (148+ packages, 70+ sales tables, 1.35B+ rows in the DICEGSA project), dimensional modeling (fact and dimension tables), incremental loads using a self-healing watermark pattern, and gold layers ready for Power BI consumption. Azure, Databricks and Microsoft Fabric ecosystem (data engineering bootcamp and a Medallion architecture proposal on Fabric).
- **Oracle (ODA) & PostgreSQL DBA:** Administration and monitoring of Oracle 19c databases on Oracle Database Appliance (RAC, Data Guard, ASM, Clusterware) for a telecommunications operator: node and instance health, Data Guard transport/apply status and lag, listeners and services, database and ASM alert logs, disk group, FRA and tablespace capacity, using odacli, srvctl, crsctl and dgmgrl. Authored a preventive-maintenance plan (daily, weekly, bi-weekly and monthly tasks). PostgreSQL administration.
- **Performance Tuning & Optimization:** Query and process optimization through execution-plan analysis (DBMS_XPLAN on Oracle, SHOWPLAN on SQL Server, EXPLAIN ANALYZE on PostgreSQL), AWR/ASH and plan stabilization. Delivered a phased roadmap and technical diagnostic documentation for the client team.
- **Database Migration:** Consolidation of 5 Oracle development environments into DB Systems on a dedicated ODA using Data Pump over network link, plus scoping of new BLOB migration requirements.
- **Business Intelligence & Analytics:** Development of BI solutions and decision-support tools with Python (Streamlit), Power BI (DAX, DirectQuery) and SQL Server: financial, sales, accounts receivable/payable and WMS dashboards.
- **Cloud & Full-Stack Architecture:** Lead the design and development of cloud-native systems and the modernization of legacy applications, eliminating on-premise dependencies. Implement high-availability architectures and event-driven solutions for critical transactions using .NET 8, Blazor WebAssembly, Azure Service Bus and Worker Services. Comprehensive infrastructure management in Azure (App Service, Azure SQL, VPN Gateway, Entra ID) ensuring security and cost optimization.
- **Cloud & FinOps Governance:** Azure infrastructure design with embedded FinOps practices: resource tagging, cost allocation per project/client, budget alerts and monthly spend reviews via Azure Cost Management. Optimized App Service tiers and Azure SQL configurations to align performance with cost targets.
- **Technical Leadership & Strategic Consulting:** Direction of development squads under Agile methodologies (Scrum). End-to-end project and bid management, including SOWs, roadmaps, infrastructure sizing and stakeholder management in highly regulated sectors (banking, government, telecommunications). Designed and delivered a 60-hour PostgreSQL 17 training plan (basic, intermediate and advanced) with a Docker lab environment.

Key Projects:

- **DICEGSA (Retail/Logistics):** Deployment of a complete Data Lake (Medallion architecture) orchestrating over 148 SSIS packages to transform 70+ sales tables, drastically reducing analytical cycle times for the BI team.
- **Ministry of Economy and Finance (MEF Panama):** End-to-end delivery of a Budgetary BI platform. Integration of a star schema in SQL Server, complex DAX metrics, DirectQuery, and an HTML landing page with role-based access control.
- **Central Bank of Nicaragua (BCN):** Architectural proposal and sizing of the SNCV system across 5 environments (Development to Production), including an open-source stack proposal (Kubernetes, Go, PostgreSQL, Keycloak), support matrix and risk classification by support tier (CapEx/OpEx) to ensure business continuity.
- **Azure + Microsoft Fabric Proposal (Banking):** Analytics platform proposal with Medallion architecture, capacity sizing (F8/F16 SKUs) and ML/AI use cases for a banking-sector entity.
- **Electoral Management System (Costa Rica):** End-to-end development of a platform to manage and control the territorial and electoral teams of a political party. Architecture built with C# .NET and Blazor, Azure DevOps pipelines, and a BI Dashboard for monitoring key metrics.
- **Pegasus System Modernization (La Costeña):** Comprehensive modernization and Azure migration, with a new web architecture on Blazor WebAssembly and .NET 8 and right-sized infrastructure with reserved-capacity planning.

Technology Solutions & Analytics Specialist
InVicta Business Partners | Remote | Aug 2025 - Jan 2026

- **Business Analysis & Process Improvement:** Strategic evaluation and implementation of technological solutions aimed at optimizing financial performance. Identification of bottlenecks and design of data-driven workflows to strengthen the operational infrastructure.
- **Data Visualization:** Design and implementation of interactive dashboards tracking critical KPIs, enabling the measurement of effectiveness and ROI for implemented solutions, while facilitating strategic decision-making.
- **Marketing Tech & Digital Analytics:** Management of web development and maintenance using CMS (WordPress, Wix Studio). Technical SEO and advanced configuration of the analytics stack (Google Analytics, Search Console, Tag Manager and Meta Pixel) to ensure accurate tracking of events and conversions.

Full Stack Architect (SaaS) & Data Analyst
Freelance / Independent Professional | Remote | Feb 2025 - Present

- **Full-Stack Development & SaaS Architecture:** Design and delivery of comprehensive, scalable and efficient web solutions (e-commerce platforms, landing pages and an ERP system in development) for international clients. Frontend (React, Next.js, Angular, TypeScript, Tailwind CSS, Zustand) and Backend (Node.js, Express, REST APIs, SQL/PostgreSQL, Supabase and Firebase).
- **E-commerce & Shopify:** Redesign and go-live of the OMG Coffee Company Shopify store (Rebel theme): brand and typography system, home page, catalog of 116 products across 14 collections, and redirection from the previous Wix site (DNS and URLs).
- **Applied AI & RAG:** Development of Extracto, an application that extracts structured data from invoices, receipts and contracts using OCR, LLMs via OpenRouter, Zod schema validation and pgvector semantic search (see Projects).
- **DevOps, CI/CD & Security:** Automation of the software lifecycle through CI/CD pipelines using GitHub Actions deploying to Vercel and Google Cloud (Cloud Run, Cloud Build). Containerized microservices with Docker, maintenance of SSISDB deployments and security audits (resolution of critical path-matching bugs in Firebase Cloud Storage).
- **Data Science & Analytics:** Advanced data analysis with Python (Pandas, NumPy, Matplotlib, Seaborn) and Power BI to turn data into strategic insight and support complex decisions; data analysis consulting for international clients.

Featured SaaS Product:

- **MiBoletoni (PaaS/SaaS):** End-to-end architecture and deployment on Google Cloud of a digital ticketing platform: payment gateways (Stripe/Fygaro), dynamic QR codes, offline-first access validation, multi-role financial administration dashboards, access control and authentication. Resolved a critical Google Cloud billing-account blockage in production, optimized Cloud Run costs (right-sized container resources) and implemented budget alerts to prevent recurrence.

Systems Analyst | Project Analyst | Software Developer
Claro Nicaragua | Digital Transformation | May 2022 - Jan 2025

- **Digital Transformation & Oracle Development:** Technical leadership in the creation and deployment of enterprise applications using Oracle SQL and Oracle APEX, automating critical operational processes for the Digital Transformation division. Design of relational data structures and contribution to the digitalization of corporate channels.
- **Cloud Adoption & Governance:** Participated in cloud operations governance, SDLC management and cost-aware infrastructure decisions for telecom-scale systems.
- **SDLC Management & Quality Control:** Full control over the software development lifecycle: architecture diagrams, schedules and resources, standardized technical documentation in Confluence and supervision of testing phases (SIT, UAT and QA) to ensure product quality.
- **Agile Methodologies & Strategic Alignment:** Coordination of cross-functional teams operating under Scrum and Kanban using Jira. Key technical liaison between business units and IT, translating requirements into clear deliverables.

Software Programmer (Oracle & PL/SQL)
Seguros Iniser | Managua, Nicaragua | Jul 2018 - Jun 2019

- **Core Development:** Evolutionary maintenance and development of information systems for the insurance sector using PL/SQL, Oracle Forms and Oracle Reports; database modeling, business logic and user interfaces built from scratch, plus technical and functional user support.
- **Data Management:** Advanced queries and administration of large volumes of transactional data, ensuring integrity, security and adherence to business logic.

Insurance Underwriting Analyst & Data Entry
Seguros Iniser | Managua, Nicaragua | Jun 2017 - Jul 2018

- **Data Analysis:** Managed and analyzed insurance policies with specialized underwriting systems and trained personnel on GPS technology for agricultural policies.

Technical Operations Assistant
RS Seguridad Digital Internacional | Managua, Nicaragua | Mar 2017 - May 2017

- **Support & Infrastructure:** Hardware maintenance, CCTV systems and on-site/remote technical support to ensure retail operational continuity.

PERSONAL & FREELANCE PROJECTS

Full portfolio: jcgadeadev.com

- **Extracto (RAG & Document Extraction):** Full-stack app (Next.js, TypeScript, PostgreSQL + pgvector, Docker) that extracts validated, structured JSON from invoices, receipts and contracts (PDF/images). OCR with Mistral OCR/Cloudflare AI, DeepSeek model via OpenRouter swappable through an environment variable, Zod validation, semantic search with embeddings, relational schema with cascading deletes and Vitest tests. Deployed on Seenode · github.com/JCGadeaDev/Extracto-RAG-DocumentExtraction
- **Proyecciones Financieras (Analítica de Datos):** Financial projections app in Python with Streamlit, Pandas, Plotly and StatsModels, published as an online demo (proyeccionesfinancieras.streamlit.app) · github.com/JCGadeaDev/ProyeccionesFinancieras_AnaliticaDatos
- **DataSet Nations (Desarrollo Humano):** Analysis and visualization of human development data with Jupyter Notebook, based on the nations.csv dataset · github.com/JCGadeaDev/DataSet_Nations
- **OMG Coffee Company (Shopify):** Production e-commerce store on Shopify (Rebel theme) with a custom brand identity, a 116-product catalog and redirection from the previous Wix site.
- **MiBoletoni:** Digital ticketing SaaS platform on Google Cloud (Next.js, Firebase, Cloud Run, Fygaro) — see Experience.
- **Panel Finanza Inteligente:** Financial dashboard (Next.js 15, TypeScript, Tailwind, Recharts, Zustand) with Google Sheets as a real-time backend, historical charts, filters and quick transaction entry.
- **Movie Vault:** Full-stack movie manager: React 19 + Node.js/Express (Render), MongoDB Atlas, Firebase Auth with Admin/Viewer roles and TMDB API.
- **Propiedades & Analytics:** Real-estate PWA with advanced search, interactive maps and price-per-square-meter analytics.
- **TechServices & Soluciones TI:** Conversion-focused corporate site with semantic SEO and load times under 1.5 s.
- **City Pizzeria:** E-commerce SPA in React 19 with centralized state, real-time cart, tax calculation and a 3-step checkout.
- **EmprenFactor:** Entrepreneurship blog (React + Vite, Firebase/Firestore, Google Search Console) deployed on Vercel.
- **Fotograma:** Personal film blog (fotograma.jcgadeadev.com) within the jcgadeadev.com site ecosystem (Astro hub on Cloudflare).

AZURE FINOPS - CORE COMPETENCY

Applying the FinOps lifecycle (Inform · Optimize · Operate) across Azure, GCP and AWS environments:

INFORM	Azure Cost Management + Billing, cost allocation by resource group/tag, telemetry health reports, custom dashboards in Power BI for spend visibility across business units.
OPTIMIZE	Right-sizing VMs and App Services, Reserved Instances negotiation, elimination of orphaned resources, Azure Advisor recommendations, storage tier optimization (Hot/Cool/Archive).
OPERATE	Budget alerts and anomaly detection, FinOps governance policies, accountability frameworks, KPI definition for long-term cost efficiency, cross-team FinOps training.
Tools	Azure Cost Management, Azure Advisor, Google Cloud Billing, AWS Cost Explorer, Power BI, Azure Monitor, Log Analytics.
Environments	Azure (primary), Google Cloud Platform (GCP), AWS - hybrid and multi-cloud architectures.
Years Active	3-5 years across Nexus Consultoría, Claro Nicaragua, Freelance/SaaS, InVicta Business Partners.

Key FinOps achievements:

- Resolved a Google Cloud billing account blockage on production SaaS (MiBoletoni/FinanzasHome), coordinating reactivation with GCP support with zero revenue loss.
- Designed cost-optimized Azure infrastructure for La Costeña (Pegasus Cloud), consolidating legacy on-premise workloads into right-sized App Service tiers with reserved capacity planning.
- Implemented resource tagging and cost allocation governance for the DICEGSA Data Lake project, enabling per-module spend tracking across Bronze/Silver/Gold layers.
- Delivered infrastructure sizing proposal for the Banco Central de Nicaragua (BCN) SNCV system across 5 environments with risk classification by support tier, directly informing CapEx/OpEx decisions.
- Managed Azure subscription cost dashboards for Nexus clients using Azure Cost Management + Power BI, producing monthly executive spend reports with actionable optimization recommendations.

EDUCATION

Master's (Continuing Education) in Big Data & Data Engineering
Complutense University of Madrid (UCM) - NTIC Master | Oct 2026 - In progress (2026/27 academic year, studied alongside full-time work)

B.S. in Systems Engineering
National University of Engineering (UNI-IES) | Managua, Nicaragua | Jul 2010 - Dec 2016 (Completed)

Business Administration (Years 1-3)
Central American University (UCA) | Managua, Nicaragua | Jan 2007 - Jun 2010

TECHNICAL SKILLS

Data Engineering & DW: Medallion Architecture (Bronze-Silver-Gold) · ETL/ELT · SSIS/SSISDB · Data Lake · Data Warehouse (Kimball/Inmon) · Data Marts · star and snowflake schemas · slowly changing dimensions (SCD) · OLAP/OLTP · Python (scripts ETL) · Azure · Databricks · Microsoft Fabric · Snowflake SQL.

Databases & DBA: Oracle Database 19c on ODA (RAC, Data Guard, ASM, Clusterware, odacli, srvctl, crsctl, dgmgrl, Data Pump, AWR/ASH, DBMS_XPLAN) · PostgreSQL (EXPLAIN ANALYZE, pgvector) · SQL Server (SSIS, SSISDB, Agent, SHOWPLAN) · MongoDB · Supabase · Firebase · query tuning and execution plans.

Data & Analytics: Power BI (DAX, DirectQuery) · Streamlit · Plotly · Python (Pandas, NumPy, Statsmodels, Matplotlib, Seaborn, Scikit-learn) · Google Analytics (GA4) · Google Tag Manager (GTM) · Looker Studio · Excel.

Languages & Runtimes: C# (.NET 8) · Python · SQL · PL/SQL · TypeScript · JavaScript (ES6+).

Backend & APIs: ASP.NET Core · Node.js · Express · Oracle APEX · Blazor Server · Razor Pages · REST APIs · Postman · SoapUI.

Frontend & UI: React · Next.js · Angular · Astro · Vite · Blazor WebAssembly · TanStack Query · Zustand · Tailwind CSS · Shadcn/UI · Bootstrap · HTML5/CSS3.

Architecture & Cloud: Azure (App Service, SQL DB, Service Bus, VPN Gateway, Entra ID, Monitor, Log Analytics, Azure DevOps Pipelines) · AWS (EC2, S3, RDS) · Google Cloud (Cloud Run, Cloud Build) · SaaS & Microservices Architecture · Event-Driven Patterns · Clean Architecture.

FinOps & Cloud Cost: Azure Cost Management · Azure Advisor · GCP Billing · AWS Cost Explorer · Reserved Instances · Right-sizing · Budget Alerts · Resource Tagging · Power BI Cost Dashboards.

DevOps & Infrastructure: Git · GitHub · GitHub Actions (CI/CD) · Docker · Terraform (IaC) · Nginx · Cloudflare · Vercel · Netlify · Render · Railway · Seenode.

AI & Agents: RAG · embeddings · OCR · OpenRouter · OpenAI API · LangChain · Claude (Anthropic) · Claude Code · Codex (OpenAI) · Gemini (Google) · DeepSeek · MCP Servers · Agent Skills · Prompt Engineering · Cursor · Zod validation · Generative AI applied to Data Engineering & Analytics.

E-commerce, CMS & Testing: Shopify · WordPress · Wix Studio · SEO Técnico · Vitest · Postman · SoapUI.

Management & Methodologies: Scrum · Kanban · Jira · Confluence · Stakeholder Management · SDLC · SOW & Bid Preparation · Design Patterns (Singleton, Repository, Factory) · Lucidchart · Miro.

CERTIFICATIONS & RELEVANT COURSEWORK

Data Engineering, SQL & Databases

DataCamp (2026): Associate Data Engineer in SQL (28 h) - Introduction to Snowflake SQL - Introduction to Python - Joining Data in SQL - Data Manipulation in SQL - Introduction to Relational Databases in SQL - Database Design - Data Warehousing Concepts - Understanding Data Engineering - Understanding Data Visualization · From Relational Model (SQL) to MongoDB's Document Model - MongoDB (2026) · credly.com/badges/d6d1a705-f849-40a7-b80c-0378835cebaf · MongoDB Aggregation Fundamentals - MongoDB (2026) · credly.com/badges/6237700e-aa5b-468d-9529-e201d92c1c7d · Data Engineering Bootcamp in Microsoft Fabric - Código Facilito (2026) · Introduction to Machine Learning - Código Facilito (2026) · Mastering SQL for Data Analysis: Extraction, Transformation & Optimization - Desafío Latam (2025) · Professional Database Course - Código Facilito (2025) · Foundations & Concepts of Data Science - Código Facilito (2025) · Introduction to Data Science - Santander Open Academy (2025) · Introduction to Data Analysis with Python - Desafío Latam (2025) · Introduction to Data Analysis with Python - Código Facilito (2025) · Power BI for Beginners - Desafío Latam (2025) · Net4Skills seminars: Powerful reports with Power BI Analytics · AWS Data Analytics Fundamentals · AWS Data Analytics (2025) · Advanced Excel - Semana de Excel (Hashtag Treinamentos) (2025)

Artificial Intelligence & AI Engineering

AI Engineer Fundamentals with Python (12-week program) - Código Facilito (2026) · Your First RAG: make AI understand your data - Código Facilito (2026) · LangChain, the AI framework for developers - Código Facilito (2026) · How to consume the OpenAI API - Código Facilito (2026) · AI Engineer Kick-off and Roadmap - Código Facilito (2026) · AI Development Fundamentals: From Zero to Production - BIG School (2026) · Exploring the Potential of Generative AI - Alura Latam (2025) · AI Immersion with Google Gemini - Alura Latam (2025) · Generative AI: automate and innovate - Netec (2025) · Webinar Azure + GenAI - Novacomp (2025)

Cloud & DevOps

Developing Solutions for Microsoft Azure - Netec (Microsoft Partner Gold Learning) (2025) · Microsoft Azure Fundamentals Course - Código Facilito (2025) · Seminar: Reduce Costs by Implementing Azure Infrastructure - Net4Skills (2025) · Terraform Specialist: Infrastructure as Code in AWS (2023) · AWS Cloud Practitioner - Centro Nacional de Tecnología (2022) · Practical AWS Course for Beginners (IAM, AMI, EC2, S3, RDS, DynamoDB) - Cloud Computing Community Nicaragua (2022) · GitHub Actions Bootcamp - Código Facilito (2025) · Professional Docker Course - Código Facilito (2025) · Multicloud Fundamentals Certificate - Braintcamp (2024) · Seminar: AWS Discovery Day - Data Analytics on AWS - Net4Skills (2025) · Seminar: VMware Cloud Foundation - Net4Skills (2025) · Secure Development Methodology Training - Security First (2023)

Web & Frontend Development

React in Action: From Zero to Expert in Front-End Development - Desafío Latam (2025) · Professional React Course - Código Facilito (2025) · Professional JavaScript Course - Código Facilito (2025) · Consuming APIs with JavaScript Fetch - Código Facilito (2025) · Responsive Web Design: HTML, CSS, Flexbox, Grid - freeCodeCamp (2022) · Introduction to Web Development I: HTML & CSS - Google Actívate (2022)

Project Management & Agile Methodologies

Foundations of Project Management - Google / Coursera (2024) · Project Management Foundations - LinkedIn Learning / PMI (2025) · Mastering JIRA for Agile Teams - LinkedIn Learning (2025) · CoderCamp Employability Program - CoderHouse (2025) · Seminar: Disciplined Agile - Net4Skills (2025)

Other Technical Training

Cisco IT Essentials: Technical Support & Operating Systems - Centro Juvenil Don Bosco (2016-2017) · Professional Networking Foundations - LinkedIn Learning (2025) · Introduction to GPON Networks - Claro Nicaragua (2023) · ISO 45001 Standard - Edu Claro (2023)

SOFT SKILLS

- **Technical Leadership:** team mentoring and strategic stakeholder management.
- **Business Acumen:** ability to translate complex business requirements into scalable technical solutions.
- **Agile Governance:** strong experience in the Software Development Life Cycle (SDLC) under Agile frameworks.
- **Problem Solving:** analytical approach to resolving complex challenges in critical environments.

LANGUAGES

- **Spanish:** Native
- **English:** B2 (Upper-Intermediate - Full Professional Proficiency)
- **Italian:** B2 (Upper-Intermediate - daily and professional use, based in Rome)